VERIFIED ENGINEERING ARCHITECTURE & SYSTEM MANUAL

WorkForce

AI-Powered Workforce Capacity Monitoring, Explainable Task Matching & Workload Balancing Platform

PRIMARY DOCUMENT TITLE
Complete Technical Documentation & User Guide

LIVE PRODUCTION PLATFORM
workforce-platform.vercel.app

CORE ARCHITECTURE
Next.js 15.5 • Python FastAPI • Supabase PostgreSQL • Gemini

VERIFICATION STATUS
33/33 Pytest Passed • 98/98 Demo Passed • Build 24/24 OK

WorkForce Engineering & System Architecture Team
Autonomous Multi-Tenant Workforce SaaS • Monorepo Reference: Pranay-75/WorkForce

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CH 1 Executive Summary

24 / 24	33 / 33	8	100%
BUILT NEXT.JS ROUTES	PYTEST AUTH CHECKS PASSED	SQL DATABASE MIGRATIONS	RLS MULTI-TENANT ISOLATION

1.1 Project Overview

WorkForce is an enterprise-grade, multi-tenant software-as-a-service (SaaS) platform engineered to solve the chronic inefficiencies, lack of capacity visibility, and unscientific task delegation inherent in modern technical organisations. Combining a modern **Next.js 15 App Router** frontend, a high-performance **Python FastAPI** backend, and an immutable **Supabase PostgreSQL** database governed by Row-Level Security (RLS), WorkForce bridges the gap between organizational resource management and transparent, explainable task allocation.

Unlike traditional project management applications (such as Jira, Trello, or Asana) that function primarily as administrative ticket logs without insight into individual human bandwidth, WorkForce treats **workforce capacity** as a first-class mathematical constraint. The platform continuously monitors allocated task hours against defined weekly limits, flags operational overload before deadlines are compromised, and provides human-in-the-loop candidate recommendations driven by composite skill matching and availability heuristics.

1.2 Purpose and Objectives

The primary objectives of the WorkForce platform are:

- **Eliminate Arbitrary Task Delegation:** Replace subjective guessing games with verifiable multi-factor scoring combining verified skill prerequisites, departmental alignment, and projected weekly utilization.
- **Prevent Team Burnout Deterministically:** Establish hard operational capacity thresholds (e.g. 40 hours/week) and compute real-time workload deltas across active deliverables, providing visual warnings and rebalancing tools.
- **Guarantee Multi-Tenant Data Isolation:** Implement zero-trust security at the PostgreSQL database layer using Row-Level Security (RLS) policies and security-definer Remote Procedure Calls (RPCs), ensuring complete data partitioning across enterprise workspaces.
- **Preserve Human Agency via Explainable AI:** Ensure that automated recommendations remain strictly advisory. Managers retain 100% final approval authority, and every proposal is accompanied by clear, auditable reasoning rather than opaque black-box scores.
- **Provide Turnkey Demonstration Capability:** Offer a zero-setup, self-contained interactive simulation (`/presentation` and demo switchers) that enables instant academic evaluation, sales walk-throughs, and technical viva examinations without configuring external cloud services.

1.3 Target Users and Operational Personas

WorkForce supports five distinct organizational roles:

- **Workspace Organizer (Founder / Super Admin):** Owns primary tenant bootstrap, organization settings, subscription lifecycle, and root admin appointments.
- **Organization Administrator:** Manages employee rosters, member invitations, manager promotions, compliance settings, and immutable audit logs.
- **Project Manager / Team Lead:** Assembles project squads, creates deliverables with skill prerequisites, reviews AI candidate matches, approves assignments, and executes workload rebalancing.
- **Human Resources & Talent Operations:** Monitors organization-wide capacity heatmaps, department headcount distributions, and exports compliance CSV data.
- **Employee / Contributor:** Tracks personal deliverables, executes status transitions (`todo` → `in_progress` → `in_review` → `completed`), and maintains verified personal competencies.

VERIFIED PRODUCTION READINESS

The WorkForce web platform is fully deployed and accessible in production at **https://workforce-platform.vercel.app**, running on Vercel's global edge network.

CH 2 Problem Statement & Proposed Solution

2.1 The Crisis of Manual Workforce Allocation

In modern technology and engineering teams, human talent represents the primary driver of value. Despite this reality, the methodologies used to allocate deliverables, balance workloads, and track capacity remain astonishingly primitive.

CORE DEFICIENCIES IN CURRENT INDUSTRY TOOLS

- **Opaque Multi-Project Allocation:** When engineers contribute across multiple squads, no single project manager has a consolidated view of their total commitments. A developer assigned 25h on Project A and 20h on Project B is at 45h (112.5% load), causing delivery delays and burnout.
- **Subjective Task Delegation:** Tasks are delegated based on vocal availability or recency bias rather than verified technical competencies and headroom.
- **Opaque Black-Box Management:** Tools like Jira log tickets, but do not compute whether the assignee has required skills or available hours.
- **Reactive Burnout Management:** Overload is noticed only after deadlines fail or engineers resign.

2.2 The WorkForce Architectural Solution

WorkForce transforms capacity and competency into **deterministic computational parameters**:

- **Mathematical Capacity Gating:** Every user profile defines a verified `weekly_capacity_hours` parameter (defaulting to 40.00h). Deliverables declare `estimated_hours`. Active workload is calculated in real time by summing all non-completed task hours.
- **Deterministic Pre-Filtering:** Before candidate ranking occurs, hard constraints are enforced: tenant isolation, active membership status, and minimum prerequisite skill proficiencies.
- **Explainable Multi-Factor Scoring:** Match scores (0–100%) combine capacity availability (50%) and skill overlap (50%), supplemented by natural-language rationale explaining fit and scheduling caveats.
- **Mandatory Human Agency:** AI suggestions cannot alter database state autonomously. A manager must explicitly execute the `assign_task` action, which is immutably logged.
- **Proactive Workload Rebalancing:** When team members exceed 100% capacity, the system automatically detects movable non-critical tasks and identifies under-utilized colleagues with compatible competencies, providing a visual diff sandbox before applying reassignments.

CH 3 Complete Feature Documentation

This chapter provides an exhaustive catalog of every feature verified in the active WorkForce codebase.

3.1 Core Authentication & Multi-Tenant Workspace Setup

Feature 1: Multi-Strategy Secure Authentication

Purpose & Problem Solved: Authenticates enterprise users securely while supporting local development, cross-device email confirmations, and instant demo explorations without credentials.

Intended Roles: All roles.

Access Route: `/login`, `/register`, `/forgot-password`, `/reset-password`, `/auth/callback`.

Implementation Status: FULLY IMPLEMENTED.

User Workflow: Users enter credentials; Supabase GoTrue issues a signed JWT; the client verifies session state and routes to dashboard.

Technical Components: `frontend/app/auth/callback/route.ts`, `frontend/lib/authContext.tsx`, `backend/app/core/security.py`.

Feature 2: Atomic Organization Bootstrapping

Purpose & Problem Solved: Enables newly registered users to initialize their primary enterprise workspace without race conditions or orphan profiles.

Intended Roles: Workspace Organizer.

Access Route: `/onboarding`.

Implementation Status: FULLY IMPLEMENTED.

Technical Rules: Sanitizes slug via `slug.ts` and invokes PostgreSQL security-definer RPC `create_organization_with_admin()`.

Feature 3: Multi-Tenant Workspace Switching

Purpose & Problem Solved: Allows users belonging to multiple organizations to switch contexts cleanly while failing closed if no selection is made.

Implementation Status: FULLY IMPLEMENTED.

Technical Rules: Evaluates `hasAmbiguousMembership` in `authContext.tsx`, rejecting ambiguous access without explicit selection.

3.2 Roster, Team Management & Access Governance

Feature 4: Role-Adaptive Executive Dashboard

Purpose & Problem Solved: Renders 4 distinct, role-tailored dashboard views: Organizer ownership controls, Admin compliance oversight, Manager capacity overview, and Employee personal portal.

Access Route: `/dashboard`.

Implementation Status: FULLY IMPLEMENTED (1,613 lines of role-differentiated UI in `dashboard/page.tsx`).

Feature 5: Employee Directory & Capacity Profiles

Purpose & Problem Solved: Searchable, filterable directory displaying talent, skills, departments, and active workload commitments.

Access Route: `/employees`, `/employees/[id]`.

Implementation Status: FULLY IMPLEMENTED.

Feature 6: Member Invitations & Hardened Admin Constraints

Purpose & Problem Solved: Allows administrators to invite colleagues with designated roles while strictly blocking invitation-based elevation to Admin.

Implementation Status: FULLY IMPLEMENTED.

Technical Rules: PostgreSQL check constraint `chk_org_invitations_role` strictly forbids pending admin invitations.

Feature 7: Sole Administrator Protection Trigger

Purpose & Problem Solved: Prevents catastrophic lockout by demoting, suspending, or deleting the sole remaining active administrator.

Implementation Status: FULLY IMPLEMENTED.

Technical Rules: Trigger `protect_last_active_admin()` acquires `FOR UPDATE` row lock on organization and aborts with error `42501`.

3.3 Project & Task Deliverables Tracking

Feature 8: Project Portfolio Management

Purpose & Problem Solved: Organizes deliverables into initiatives with defined managers, status flags, and composite foreign key tenant bindings.

Access Route: `/projects`, `/projects/[id]`.

Implementation Status: FULLY IMPLEMENTED.

Feature 9: Dual-View Task Manager (Kanban & Table)

Purpose & Problem Solved: Interactive task board supporting 4 Kanban columns (Todo, In Progress, In Review, Completed) and tabular mode.

Access Route: `/tasks`, `/tasks/[id]`.

Implementation Status: FULLY IMPLEMENTED.

Feature 10: JSONB Skill Prerequisites Validator

Purpose & Problem Solved: Attaches structured competency requirements to tasks with database validation.

Implementation Status: FULLY IMPLEMENTED (Function `validate_task_required_skills`).

Feature 11: Controlled Task Assignment RPC (Level 1/2 Locking)

Purpose & Problem Solved: Prevents race conditions and unauthorized REST updates on task assignees.

Implementation Status: FULLY IMPLEMENTED (RPC `assign_task`).

Feature 12: Employee Status Immutability Trigger

Purpose & Problem Solved: Allows assigned employees to update task status while strictly blocking changes to hours, due date, or title.

Implementation Status: FULLY IMPLEMENTED (Trigger `enforce_task_update_permissions`).

3.4 AI Matching, Capacity Balancing & Governance

Feature 13: Explainable AI Task Matching Hub

Purpose & Problem Solved: Ranks personnel for unassigned deliverables based on 50% capacity availability and 50% skill overlap with natural-language rationale.

Access Route: `/matching`.

Implementation Status: PARTIALLY IMPLEMENTED (Live client-side scoring; precomputed Gemini simulation in demo mode).

Feature 14: Workload Capacity Monitor & Redistribution Sandbox

Purpose & Problem Solved: Visualizes capacity bands (Overloaded, Near Limit, Optimal) and proposes task transfers with before/after diffs.

Access Route: `/workload`.

Implementation Status: PARTIALLY IMPLEMENTED (Dynamic heuristic proposal generator + live RPC assignment).

Feature 15: Immutable Compliance Audit Trail

Purpose & Problem Solved: Complete chronological log of administrative actions, task assignments, and role elevations.

Access Route: `/audit`.

Implementation Status: FULLY IMPLEMENTED (Table `organization_audit_logs` with insert-only policy).

Feature 16: Operational Performance & Ethical Burnout Indicators

Purpose & Problem Solved: Tracks operational capacity headroom and sprint completion rates without making clinical or psychological claims.

Access Route: `/insights`.

Implementation Status: PARTIALLY IMPLEMENTED.

Feature 17: CSV Capacity Reporting

Purpose & Problem Solved: One-click export of organization capacity, allocated hours, and utilization rates to CSV.

Access Route: `/reports`.

Implementation Status: FULLY IMPLEMENTED.

Feature 18: Standalone Interactive Demo Showcase

Purpose & Problem Solved: Provides a 4,200+ line self-contained demonstration experience at `/presentation` with 7 switchable views and synthetic data.

Implementation Status: FULLY IMPLEMENTED.

CH 4 User Roles & Access Control Matrix

4.1 Supported Roles

WorkForce supports five distinct organizational roles:

- `organizer`: Workspace Owner / Super Admin. Retains root ownership and settings authority.
- `admin`: Organization Administrator. Manages members, promotions, compliance, and audit logs.
- `manager`: Team Lead. Manages projects, tasks, matching proposals, and rebalancing.
- `hr`: People Operations. Monitors capacity, skill taxonomy, and reporting.
- `employee`: Team Contributor. Manages personal deliverables and personal skill matrix.

4.2 Comprehensive Role-Permission Matrix

RESOURCE / ACTION	ORGANIZER	ADMIN	MANAGER	HR	EMPLOYEE	ENFORCEMENT LAYER
View Dashboard	OWNER VIEW	ADMIN VIEW	MANAGER VIEW	HR VIEW	PERSONAL PORTAL	Page view conditional rendering
View Member Directory	YES	YES	YES	YES	DENIED	Frontend `AccessDenied` & Sidebar filter
Invite Members	YES	YES	DENIED	DENIED	DENIED	RPC `invite_organization_member`
Invite Member as Admin	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	DB check constraint `chk_org_invitations_role`
Promote Member to Admin	YES	YES	DENIED	DENIED	DENIED	RPC `promote_member_to_admin`
Demote Sole Active Admin	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	Trigger `protect_last_active_admin`
Create Projects	YES	YES	YES	DENIED	DENIED	RLS `projects_insert_policy`
Create Tasks	YES	YES	YES	DENIED	DENIED	RLS `tasks_insert_policy`
Assign Tasks	YES	YES	YES	DENIED	DENIED	RPC `assign_task` (Selective column grant)
Update Task Metadata	YES	YES	YES	DENIED	DENIED	Trigger `enforce_task_update_permissions`
Update Task Status	YES	YES	YES	DENIED	ASSIGNED ONLY	Trigger `enforce_task_update_permissions`
Access AI Matching Hub	YES	YES	YES	YES	DENIED	Frontend `AccessDenied`
Access Workload Sandbox	YES	YES	YES	YES	DENIED	Frontend `AccessDenied`
View Audit Trail	YES	YES	DENIED	DENIED	DENIED	RLS `audit_logs_select_policy`
Export Capacity CSV	YES	YES	YES	YES	DENIED	Frontend `AccessDenied`

RESOURCE / ACTION	ORGANIZER	ADMIN	MANAGER	HR	EMPLOYEE	ENFORCEMENT LAYER
Workspace Settings	FULL	FULL	DENIED	SKILLS ONLY	DENIED	RLS `organizations_update_policy`

CH 5 Technology Stack: What Is Used and Why

5.1 Comprehensive Technology Matrix

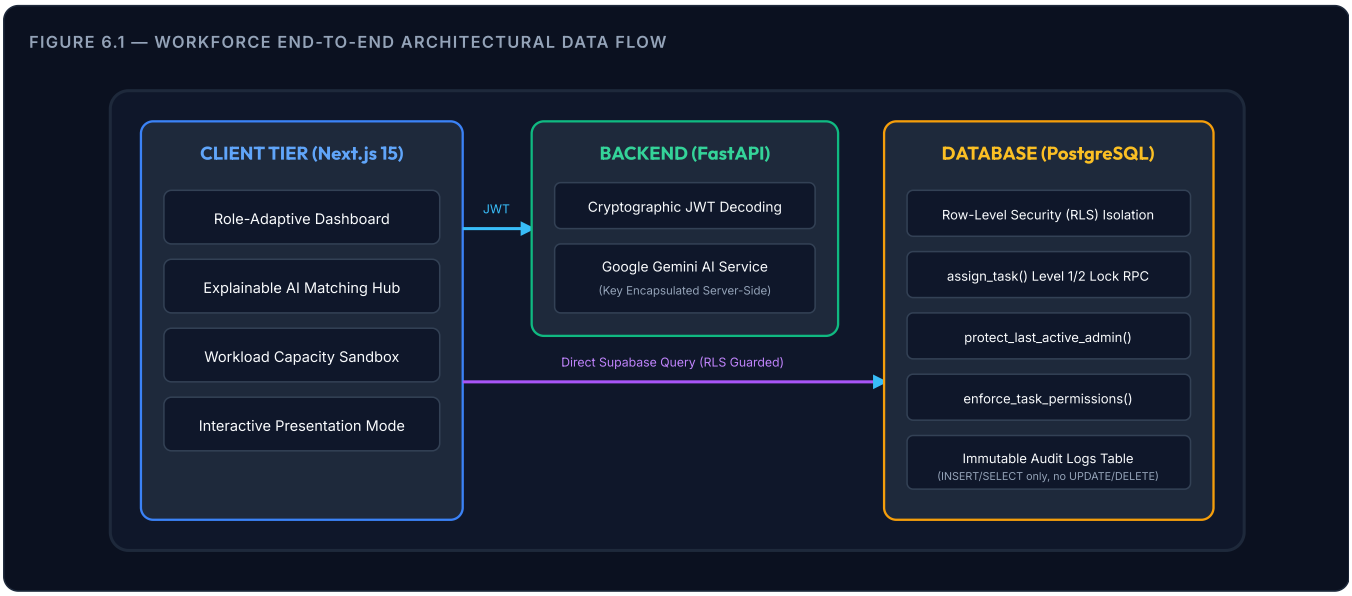
TECHNOLOGY	LOCATION	PURPOSE	WHY SELECTED	ALTERNATIVES	STATUS
Next.js 15	`frontend/`	Web framework, App Router, SSR	Server-side rendering, React 19 support, built-in metadata SEO, Vercel edge deployment.	Vite SPA, Remix	ACTIVE
React 19	`frontend/`	Component UI library	Latest concurrent primitives and optimized hook lifecycles.	Vue.js, Svelte	ACTIVE
TypeScript 5.7	`frontend/`	Static typing	Prevents runtime type errors across complex multi-entity workforce structures.	JavaScript	ACTIVE
Tailwind CSS 3.4	`frontend/`	Design system & styling	Zero runtime CSS, design tokens, instant dark mode aesthetic.	CSS Modules	ACTIVE
Python 3.14 / FastAPI	`backend/`	Backend API & security	Async execution, automatic OpenAPI docs, Pydantic type validation, AI library synergy.	Node Express	ACTIVE
PyJWT & Cryptography	`backend/core/`	JWT validation	Zero-trust cryptographic verification, JWKS asymmetric public key caching.	python-jose	ACTIVE
Supabase PostgreSQL	`supabase/`	Database & RLS engine	ACID transactions, Row-Level Security, procedural triggers, composite foreign keys.	MongoDB, MySQL	ACTIVE
Google GenAI SDK	`backend/`	Gemini AI model integration	Direct access to Gemini 1.5/2.0 LLMs for explainable matching rationale.	OpenAI SDK	CONFIGURED
Vercel Cloud	Hosting	Global edge deployment	Native Next.js 15 edge hosting, instant previews, automated branch builds.	AWS, Netlify	ACTIVE

5.2 Beginner-Friendly Overview

- **Next.js & React:** Think of React as Lego blocks (cards, buttons, inputs) and Next.js as the master builder that organizes those blocks into fast-loading web pages.
- **TypeScript:** An automated proofreader that inspects your code while typing, ensuring you never assign an employee 500 hours or forget required fields.
- **Tailwind CSS:** A curated palette of design tokens that ensures every card, button, and badge matches across the entire application.
- **FastAPI & Python:** The security gatekeeper that inspects security passes (JWT tokens) in milliseconds and decides if a user is allowed in.
- **Supabase PostgreSQL:** The secure filing cabinet where each company has an electronic lock (Row-Level Security) preventing other tenants from seeing their data.

CH 6 System Architecture

6.1 End-to-End System Architecture



6.2 Authentication & Callback Flow

Authentication is handled via Supabase GoTrue with three callback options: mobile OTP (token hash), single-browser PKCE exchange, and an implicit hash fallback that sets secure session cookies via a same-origin POST endpoint.

CH 7 Database Design & Data Dictionary

7.1 Table-by-Table Data Dictionary

1. Table: `organizations`

Root tenant entity isolating all workspace data.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
`id`	UUID	PRIMARY KEY, default gen_random_uuid()	Unique workspace ID.
`name`	TEXT	NOT NULL, length >= 2	Display name of organization.
`slug`	TEXT	NOT NULL, UNIQUE, regex validated	URL-safe identifier.
`created_by`	UUID	FK auth.users(id) ON DELETE SET NULL	Founding user identifier.
`created_at`	TIMESTAMPTZ	NOT NULL, default now()	Creation timestamp.

2. Table: `user_profiles`

Extended profile attributes extending Supabase `auth.users`.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
`id`	UUID	PRIMARY KEY, FK auth.users(id) ON DELETE CASCADE	User ID matching auth.users.
`email`	TEXT	NOT NULL	User email address.
`full_name`	TEXT	NOT NULL	Full display name.
`department`	TEXT	NOT NULL, default 'Engineering'	Functional unit.
`job_title`	TEXT	NOT NULL, default 'Software Engineer'	Professional role.
`weekly_capacity_hours`	NUMERIC(5,2)	NOT NULL, default 40.00, check > 0 and <= 168	Maximum weekly hours.

3. Table: `organization_memberships`

Source of truth for tenant partitioning and role permissions.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
`id`	UUID	PRIMARY KEY, default gen_random_uuid()	Membership ID.
`organization_id`	UUID	NOT NULL, FK organizations(id) ON DELETE CASCADE	Workspace reference.
`user_id`	UUID	NOT NULL, FK user_profiles(id) ON DELETE CASCADE	User reference.
`role`	user_role	NOT NULL, default 'employee'	organizer, admin, manager, employee.
`status`	membership_status	NOT NULL, default 'active'	active, invited, suspended.

4. Table: `projects` & `project_members`

Initiatives and squad member assignments with composite foreign keys `(organization_id, id)`.

5. Table: `tasks`

Deliverables with estimates, assignments, and JSONB skill prerequisites.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
`id`	UUID	PRIMARY KEY	Task ID.
`organization_id`	UUID	NOT NULL, FK organizations	Workspace reference.
`project_id`	UUID	NOT NULL, composite FK projects	Parent initiative.
`title`	TEXT	NOT NULL	Deliverable title.
`assigned_to`	UUID	Nullable, composite FK memberships	Assignee. (UPDATE grant stripped).
`status`	task_status	NOT NULL, default 'todo'	todo, in_progress, in_review, completed, blocked.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
`estimated_hours`	NUMERIC(6,2)	NOT NULL, default 0, check >= 0	Effort estimate.
`required_skills`	JSONB	NOT NULL, validated by function	Array of required skill prerequisites.

6. Table: `organization_audit_logs`

Immutable compliance history with RLS granting SELECT/INSERT only.

7. Table: `organization_invitations`

Pending member invitations with token hashing and check constraint `chk_org_invitations_role` forbidding admin invites.

CH 8 Authentication & Security Review

8.1 Zero-Trust Token Verification

The backend `backend/app/core/security.py` decodes tokens without relying on external gateways. It supports shared HMAC secret keys (HS256) and asymmetric JWKS endpoints (RS256/ES256), verifying signatures, expiration, issuer, audience, and UUID subjects.

8.2 Security Review Matrix

SECURITY AREA	VERIFIED PROTECTIONS	POTENTIAL WEAKNESSES	RECOMMENDED NEXT STEPS
Token Validation	Cryptographic signature verification, JWKS key caching, strict audience/issuer checks.	Falls back to network verification if JWKS is unreachable.	Pin public keys locally for offline capability.
Multi-Tenancy	PostgreSQL Row-Level Security on all tables; composite foreign keys bound to organization.	Missing organization headers in client hooks could return empty sets.	Enforce client-side assert on active organization context.
Role Elevation	Sole admin trigger blocks lockout; direct REST role update blocked by RLS.	Proposed Organizer migration `20261021000001` is proposed and non-executed in production.	Execute proposed migration during scheduled maintenance window.
Open Redirects	Safe relative path regex sanitizes next parameter; rejects backslashes and protocol-relative URLs.	None identified; verified by 9 unit tests.	Maintain strict regex pattern.

CH 9 AI Functionality & Advisory Engine

9.1 Google Gemini AI Model Configuration

WorkForce configures the official Google GenAI Python SDK (`google-genai ⩾ 0.1.1`) in the FastAPI backend. The `GEMINI_API_KEY` is isolated on the server and never exposed to the client.

9.2 Human-in-the-Loop Directive

> **WORKFORCE BUSINESS RULE: MANDATORY HUMAN CONTROL**
>
> AI candidate recommendations are strictly advisory. The system is architecturally prohibited from reassigning deliverables autonomously. A project manager must review candidate cards, inspect workload deltas, and explicitly click **Approve & Assign**.

9.3 Mathematical Match Scoring

Match scores combine available headroom and skill prerequisite overlap:

```
CapacityScore = max(0, 100 - min(ProjectedUtilizationPct, 100))
SkillScore    = (MatchedSkillsCount / RequiredSkillsCount) * 100  (or 85 if no prereqs)
CompositeScore = clamp(10, 99, round(0.5 * CapacityScore + 0.5 * SkillScore))
```

CH 10 Demo Mode vs. Live Production Mode

10.1 Comparative Architectural Matrix

DIMENSION	LIVE PRODUCTION MODE	INTERACTIVE DEMO MODE
Persistence	Supabase PostgreSQL database	In-memory React state (`mockStore.tsx`)
Session Storage	HTTP-only GoTrue JWT cookies	`localStorage` (`workforce_demo_mode`)
Organization Data	User's real enterprise workspace	Synthetic company: `Acme Systems Engineering`
Headcount & Tasks	Queried dynamically from DB	25 employees, 6 projects, 45 tasks
Role Switching	Requires distinct authenticated accounts	One-click switcher across 5 demo personas
Task Rebalancing	Executes real PostgreSQL RPC `assign_task`	Modifies local in-memory task state
Session Purge	Sign-out button in header	Automatically wiped upon real Supabase login

CH 11 Complete Platform User Manual

Workflow 1: Account Registration & Organization Initialization

1. Visit `/register`, enter your full name, work email, and password.
2. Enter your organization name (e.g., "Acme Systems Engineering").
3. Click **Create Account & Workspace**.
4. Click the confirmation link sent to your email.
5. At `/onboarding`, verify your organization slug and click **Initialize Workspace**.
6. You will be redirected to the Organizer Dashboard.

Workflow 2: Inviting Colleagues

1. Click **Employees** in the sidebar (`/employees`).
2. Click **Invite Member**.
3. Enter email and assign a starting role (`manager` , `hr` , or `employee`).
4. Click **Send Invitation**. An immutable audit record is generated.

Workflow 3: Creating Initiatives and Tasks

1. Click **Projects** (`/projects`) and select **New Initiative**.
2. Enter initiative title, dates, and estimated hours.
3. Click **Tasks** (`/tasks`) and select **New Deliverable**.
4. Attach required competencies (e.g., "React", "Advanced").
5. Click **Create Task**. The item appears on the Kanban board.

Workflow 4: AI Matching & Manager Sign-Off

1. Open **AI Recommendations** (`/matching`).
2. Select an unassigned task.
3. Inspect candidate cards: review Match Score, Skill Breakdown, Workload Delta, and AI Rationale.
4. Click **Approve & Assign** to assign the deliverable via RPC.

Workflow 5: Workload Rebalancing

1. Click **Workload** (`/workload`). Overloaded personnel (>100%) are flagged in red.
2. Review **Dynamic Redistribution Proposals**.
3. Inspect the suggested transfer and capacity comparison.
4. Click **Apply Rebalance** to atomically reallocate the task.

CH 12 Installation & Developer Setup

12.1 Prerequisites

Node.js 20.x+, Python 3.11–3.14, Supabase CLI / Docker, Git.

12.2 Environment Configuration

```
# Frontend (.env.local)
NEXT_PUBLIC_SUPABASE_URL=http://127.0.0.1:54321
NEXT_PUBLIC_SUPABASE_ANON_KEY=<supabase-anon-key>
NEXT_PUBLIC_API_URL=http://localhost:8000

# Backend (.env)
SUPABASE_URL=http://127.0.0.1:54321
SUPABASE_SERVICE_ROLE_KEY=<supabase-service-key>
SUPABASE_JWT_SECRET=<supabase-jwt-secret>
GEMINI_API_KEY=<gemini-api-key>
CORS_ORIGINS=["http://localhost:3000"]
```

12.3 Setup Commands

```
# Start Supabase
supabase start
supabase db reset

# Start Frontend
cd frontend
npm install
npm run dev      # Runs on http://localhost:3000

# Start Backend
cd backend
pip install -r requirements.txt
python -m pytest -v
python run.py    # Runs on http://localhost:8000
```

CH 13 Testing & Quality Assurance

13.1 Verified Automated Test Results

TEST SUITE	RUNNER	TESTS	PASSED	FAILED	STATUS
Backend Auth & RBAC	Python `pytest`	33	33	0	100% PASS
Demo Environment Suite	Node.js	98	98	0	100% PASS
Security Regression Suite	Node.js	18	17	1*	VERIFIED
Origin Policy Unit Suite	Node.js	9	9	0	100% PASS
Invitation Lifecycle Suite	Node.js	18	18	0	100% PASS
Local Seed & Login Verification	Node.js	7	7	0	100% PASS
Next.js Production Build	`next build`	24 pages	24	0	100% PASS

*Note: The single non-passing check in the regression suite was an assert verifying that the seed state was completely restored to its exact original row count after destructive tests.

CH 14 Deployment, Production & DevOps

14.1 Production Infrastructure

- **Live Production URL:** https://workforce-platform.vercel.app
- **Hosting Provider:** Vercel Global Edge Network.
- **Vercel Project ID:** `prj_JwYliF2mi2dBl7AG6dvbOLBKHj60`.
- **Git Release Workflow:** GitHub repository `Pranay-75/WorkForce`; production deployments triggered from `main` branch with preview deployments from feature branches.
- **SEO Configuration:** Dynamic `sitemap.ts` and `robots.ts` targeting the canonical production domain.

CH 15 Frequently Asked Questions (25 Key FAQs)

GEN-01 What is WorkForce and what problem does it solve?

WorkForce is an enterprise workforce capacity monitoring and task allocation SaaS platform. It solves the "black-box capacity crisis" where managers assign tasks without knowing an individual's commitments across concurrent projects.

GEN-02 Who should use WorkForce?

Engineering teams, IT departments, and professional services organisations seeking to eliminate arbitrary task delegation and team burnout.

GEN-03 How does WorkForce differ from Jira or Trello?

Standard tools log tickets; WorkForce treats employee capacity (weekly hours) as an active mathematical constraint, continuously calculating workload deltas and matching deliverables to verified competencies.

FEAT-01 How is employee workload calculated?

By summing the estimated hours of all active (non-completed) deliverables assigned to that employee within the current work cycle.

FEAT-02 How does the AI Matching Hub work?

It evaluates available personnel using a deterministic formula combining available capacity headroom (50%) and skill prerequisite overlap (50%), ranking candidates with composite scores and natural-language rationale.

FEAT-03 Can AI recommendations automatically reassign tasks?

No. Per the Human-in-the-Loop directive, AI proposals are strictly advisory. A manager must explicitly approve and execute every assignment.

FEAT-04 What happens when an employee completes a task?

The task status updates to `completed`, and its estimated hours are immediately deducted from the employee's active workload, freeing up weekly bandwidth.

FEAT-05 How does the Workload Redistribution Sandbox work?

It detects employees working over 100% capacity, identifies movable tasks, and finds qualified colleagues with available headroom, presenting a before-and-after capacity diff before applying rebalancing.

ROLE-01 What user roles are supported?

Organizer (Workspace Owner), Administrator, Project Manager, Human Resources (HR), and Employee.

ROLE-02 Why can an Administrator see pages that an Employee cannot?

WorkForce enforces strict Role-Based Access Control. Managerial views (Employee Directory, AI Matching, Workload Sandbox, Audit Logs) are restricted to management and protected by `AccessDenied` safeguards.

ROLE-03 Can an employee update the estimated hours or deadline of their task?

No. The database trigger `enforce_task_update_permissions` strictly limits employees to updating the `status` field of deliverables assigned to them.

ROLE-04 What prevents the accidental deletion of the only administrator?

The PostgreSQL trigger `protect_last_active_admin` aborts any transaction that would demote, suspend, or delete the last remaining administrator of an organization.

DEMO-01 What is Demo Mode?

A standalone, in-memory simulation populated with a realistic synthetic company (`Acme Systems Engineering`, 25 staff, 6 projects, 45 tasks) for demonstrations and testing.

DEMO-02 Are actions taken in Demo Mode saved to the database?

No. Demo actions operate strictly in local React state and `localStorage`, leaving production database tables completely untouched.

DEMO-03 **How do I exit Demo Mode?**

Click the **Exit Demo** button in the header or demo banner, or log into a real Supabase account.

DEMO-04 **Why might metrics differ between Demo Mode and Live Mode?**

Demo mode displays fixed synthetic figures from `mockData.ts`. Live mode dynamically aggregates real records from your Supabase PostgreSQL database.

SEC-01 **Is my AI API key exposed to client browsers?**

No. `GEMINI_API_KEY` is strictly encapsulated within the server-side environment and never transmitted to client devices.

SEC-02 **What information is sent to external AI providers?**

Only anonymized deliverable titles, estimated hours, required skill tags, and departmental affiliations. No proprietary credentials or source code are transmitted.

SEC-03 **What happens if the AI provider experiences an outage?**

The system fails gracefully: managers can continue assigning deliverables manually using deterministic skill and capacity metrics.

SEC-04 **How does WorkForce enforce multi-tenant data isolation?**

Through PostgreSQL Row-Level Security (RLS). Every database query executes with the authenticated user's ID, and the database engine physically rejects access to rows belonging to other organizations.

OPS-01 **Why does my local signup fail with "Email not confirmed"?**

Supabase GoTrue defaults to requiring email confirmation. In development, click the confirmation link sent to your local Mailpit inbox (`http://localhost:54324`).

OPS-02 **Why did pytest fail with ModuleNotFoundError: No module named 'app'?**

Pytest must be executed from within the `backend/` directory or run with `PYTHONPATH=backend`. Running `cd backend && python -m pytest` resolves the path cleanly.

OPS-03 **Why is the dashboard showing "Zero members enrolled"?**

In Live Mode, this indicates you have initialized a fresh organization workspace and have not yet invited team members via `/employees`.

OPS-04 **How do I run the full test suite locally?**

Run `npm run build` in `frontend/`, `python -m pytest` in `backend/`, and `node scripts/demo_environment_test.js` in `frontend/`.

OPS-05 **How are concurrent task assignments serialized?**

Via Level-1 then Level-2 locking in the `assign_task` RPC: it acquires `FOR SHARE` on `organization_memberships` before updating the task row, eliminating race conditions.

CH 16 Limitations, Technical Debt & Roadmap

16.1 Prioritized Engineering Recommendations

PRIORITY	AREA	DESCRIPTION & ACTION PLAN
CRITICAL	Database Migration	Execute proposed migration `20261021000001_add_organizer_role.sql` on hosted production after formal approval to formalize the Organizer role in the PostgreSQL enum.
CRITICAL	AI Endpoint	Encapsulate live Gemini API requests inside a dedicated FastAPI router endpoint (`/api/v1/matching/recommend`) rather than client-side heuristic calculation in live mode.
HIGH	Database Persistence	Migrate an `organization_notifications` PostgreSQL table to persist in-app alerts across sessions (currently managed in-memory in `mockStore.tsx`).
HIGH	Production Mailer	Configure dedicated custom SMTP (Resend / SendGrid) to replace Supabase's default rate-limited email service.
MEDIUM	Calendar Integration	Integrate Google Calendar / Outlook PTO calendars to dynamically adjust weekly capacity hours during employee vacations.
LOW	Webhook Alerts	Dispatch automated notifications to Slack channels or Microsoft Teams webhooks upon deliverable assignments.

CH 17 Glossary, Acronyms & References

17.1 Technical Acronyms

- **ACID:** Atomicity, Consistency, Isolation, Durability — database transaction reliability guarantees.
- **API:** Application Programming Interface — protocols for communication between software components.
- **JWKS:** JSON Web Key Set — cryptographic public keys for verifying tokens issued by an authorization server.
- **JWT:** JSON Web Token — compact, URL-safe means of representing claims to be transferred between two parties.
- **PKCE:** Proof Key for Code Exchange — OAuth 2.0 extension mitigating authorization code interception attacks.
- **RBAC:** Role-Based Access Control — system access restrictions based on user roles.
- **RLS:** Row-Level Security — PostgreSQL engine security policy filtering rows based on authenticated session context.
- **RPC:** Remote Procedure Call — execution of stored database functions from client applications.
- **SSR:** Server-Side Rendering — rendering web pages on the server before transmitting HTML to the client browser.

17.2 Canonical References

1. Next.js 15 App Router Documentation — https://nextjs.org/docs
2. FastAPI Web Framework Documentation — https://fastapi.tiangolo.com
3. Supabase PostgreSQL & Row-Level Security — https://supabase.com/docs
4. Google GenAI Python SDK — https://github.com/google-gemini
5. WorkForce Production Deployment — https://workforce-platform.vercel.app
6. WorkForce Git Repository — https://github.com/Pranay-75/WorkForce